THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(b) $\frac{1}{6}$	There are 36 equally likely outcomes.
2	(c) $\frac{1}{6}$	Count favourable outcomes out of 36.
3	(c) $\frac{2}{3}$	Count the outcomes in either event, without double counting.
4	(c) $\frac{7}{12}$	Both favourable and total counts increase.
5	(d) $\frac{4}{11}$	Count the matching letters, including repeats.
6	(d) 7200	Expected number = total × probability.
7	(b) $\frac{7}{12}$	Use the complement rule.
8	(a) $\frac{1}{2}$	Use the 8 outcomes.
9	(d) $\frac{1}{9}$	There are 36 equally likely outcomes.
10	(b) $\frac{1}{6}$	Count favourable outcomes out of 36.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Outcomes with sum 6: (1,5), (2,4), (3,3), (4,2), (5,1) (1); $P = \frac{5}{36} = \frac{5}{36}$ (1).	1+1
12	New red = 9, new total = 18 (1); $P = \frac{1}{2}$ (1).	1+1
13	$P(\text{not win}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1); Expected wins = $200 \times \frac{2}{5} = 80$ (1).	1+1
14	Outcomes with sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) (1); $P = \frac{5}{36} = \frac{5}{36}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{a number less than } 3) = \frac{1}{3}$ (1); $P(\text{a number greater than } 4) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
16	Likes tea = 70 + 30 = 100; $P(\text{tea}) = \frac{100}{200} = \frac{1}{2}$ (1); Neither = 200 − 70 − 50 − 30 = 50 (1); $P(\text{neither}) = \frac{1}{4}$ (1).	1+1+1
17	Let x red balls be added: $\frac{3+x}{12+x} = \frac{1}{2}$ (1); $2(3+x) = 1(12+x)$ (1); $x = 6$ (1).	1+1+1
	OR: $P(\text{a prime number}) = \frac{1}{2}$ (1); $P(\text{a multiple of } 3) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
18	Likes tea = 110 + 40 = 150; $P(\text{tea}) = \frac{150}{300} = \frac{1}{2}$ (1); Neither = 300 − 110 − 60 − 40 = 90 (1); $P(\text{neither}) = \frac{3}{10}$ (1).	1+1+1
	OR: Let x red balls be added: $\frac{4+x}{12+x} = \frac{2}{3}$ (1); $3(4+x) = 2(12+x)$ (1); $x = 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum 5: 4 outcomes; sum 9: 4 outcomes (1); $P(5) = \frac{1}{9}$ (1); $P(9) = \frac{1}{9}$ (1); $P(\text{either}) = \frac{8}{36} = \frac{2}{9}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{8}{200} = \frac{1}{25}$ (1); Expected defective = $5000 \times \frac{1}{25}$ (1); = 200 (1); Units stated (2).	1+1+1+2
20	Sum 4: 3 outcomes; sum 10: 3 outcomes (1); $P(4) = \frac{1}{12}$ (1); $P(10) = \frac{1}{12}$ (1); $P(\text{either}) = \frac{6}{36} = \frac{1}{6}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{15}{500} = \frac{3}{100}$ (1); Expected defective = $20000 \times \frac{3}{100}$ (1); = 600 (1); Units stated (2).	1+1+1+2

Mistakes to avoid:

- Giving a probability greater than 1 or a negative value, or leaving it as a count instead of a fraction.
- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(c) $\frac{2}{3}$	Count the outcomes in either event, without double counting.
2	(b) $\frac{5}{8}$	Both favourable and total counts increase.
3	(d) $\frac{4}{11}$	Count the matching letters, including repeats.
4	(c) 5600	Expected number = total × probability.
5	(a) $\frac{8}{15}$	Use the complement rule.
6	(c) $\frac{1}{4}$	Use the 8 outcomes.
7	(d) $\frac{1}{12}$	There are 36 equally likely outcomes.
8	(c) $\frac{1}{36}$	Count favourable outcomes out of 36.
9	(d) $\frac{2}{3}$	Count the outcomes in either event, without double counting.
10	(d) $\frac{8}{15}$	Both favourable and total counts increase.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New red = 10, new total = 20 (1); $P = \frac{1}{2}$ (1).	1+1
12	$P(\text{not win}) = 1 - \frac{3}{8} = \frac{5}{8}$ (1); Expected wins = $160 \times \frac{3}{8} = 60$ (1).	1+1
13	Outcomes with sum 10: (4,6), (5,5), (6,4) (1); $P = \frac{3}{36} = \frac{1}{12}$ (1).	1+1
14	New red = 9, new total = 14 (1); $P = \frac{9}{14}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{an even number}) = \frac{1}{2}$ (1); $P(\text{a multiple of } 3) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
16	Likes tea = $40 + 20 = 60$; $P(\text{tea}) = \frac{60}{120} = \frac{1}{2}$ (1); Neither = $120 - 40 - 30 - 20 = 30$ (1); $P(\text{neither}) = \frac{1}{4}$ (1).	1+1+1
17	Let x red balls be added: $\frac{5+x}{20+x} = \frac{3}{4}$ (1); $4(5+x) = 3(20+x)$ (1); $x = 40$ (1).	1+1+1
	OR: $P(\text{a perfect square}) = \frac{1}{3}$ (1); $P(\text{a number at least } 5) = \frac{1}{3}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
18	Likes tea = $90 + 35 = 125$; $P(\text{tea}) = \frac{125}{250} = \frac{1}{2}$ (1); Neither = $250 - 90 - 70 - 35 = 55$ (1); $P(\text{neither}) = \frac{11}{50}$ (1).	1+1+1
	OR: Let x red balls be added: $\frac{2+x}{12+x} = \frac{1}{2}$ (1); $2(2+x) = 1(12+x)$ (1); $x = 8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum 6: 5 outcomes; sum 8: 5 outcomes (1); $P(6) = \frac{5}{36}$ (1); $P(8) = \frac{5}{36}$ (1); $P(\text{either}) = \frac{10}{36} = \frac{5}{18}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{10}{250} = \frac{1}{25}$ (1); Expected defective = $7500 \times \frac{1}{25}$ (1); = 300 (1); Units stated (2).	1+1+1+2
20	Sum 2: 1 outcomes; sum 12: 1 outcomes (1); $P(2) = \frac{1}{36}$ (1); $P(12) = \frac{1}{36}$ (1); $P(\text{either}) = \frac{2}{36} = \frac{1}{18}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{12}{400} = \frac{3}{100}$ (1); Expected defective = $16000 \times \frac{3}{100}$ (1); = 480 (1); Units stated (2).	1+1+1+2

Mistakes to avoid:

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- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{6}{11}$	Count the matching letters, including repeats.
2	**(d)** 4000	Expected number = total × probability.
3	**(b)** $\frac{9}{20}$	Use the complement rule.
4	**(c)** $\frac{7}{8}$	Use the 8 outcomes.
5	**(a)** $\frac{1}{18}$	There are 36 equally likely outcomes.
6	**(b)** $\frac{1}{2}$	Count favourable outcomes out of 36.
7	**(b)** $\frac{2}{3}$	Count the outcomes in either event, without double counting.
8	**(a)** $\frac{7}{15}$	Both favourable and total counts increase.
9	**(a)** $\frac{4}{11}$	Count the matching letters, including repeats.
10	**(c)** 12000	Expected number = total × probability.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$P(\text{not win}) = 1 - \frac{5}{12} = \frac{7}{12}$ (1); Expected wins $= 240 \times \frac{5}{12} = 100$ (1).	1+1
12	Outcomes with sum 3: (1,2), (2,1) (1); $P = \frac{2}{36} = \frac{1}{18}$ (1).	1+1
13	New red = 15, new total = 21 (1); $P = \frac{5}{7}$ (1).	1+1
14	$P(\text{not win}) = 1 - \frac{4}{9} = \frac{5}{9}$ (1); Expected wins $= 360 \times \frac{4}{9} = 160$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$P(\text{a factor of } 6) = \frac{2}{3}$ (1); $P(\text{a composite number}) = \frac{1}{3}$ (1); Favourable for either: 5, so $P = \frac{5}{6}$ (1).	1+1+1
16	Likes tea $= 150 + 60 = 210$; $P(\text{tea}) = \frac{210}{400} = \frac{21}{40}$ (1); Neither $= 400 - 150 - 100 - 60 = 90$ (1); $P(\text{neither}) = \frac{9}{40}$ (1).	1+1+1
17	Let x red balls be added: $\frac{6+x}{18+x} = \frac{3}{4}$ (1); $4(6+x) = 3(18+x)$ (1); $x = 30$ (1).	1+1+1
	OR: $P(\text{a number not divisible by } 3) = \frac{2}{3}$ (1); $P(\text{the number } 5) = \frac{1}{6}$ (1); Favourable for either: 4, so $P = \frac{2}{3}$ (1).	1+1+1
18	Likes tea $= 55 + 25 = 80$; $P(\text{tea}) = \frac{80}{160} = \frac{1}{2}$ (1); Neither $= 160 - 55 - 40 - 25 = 40$ (1); $P(\text{neither}) = \frac{1}{4}$ (1).	1+1+1
	OR: Let x red balls be added: $\frac{4+x}{20+x} = \frac{3}{5}$ (1); $5(4+x) = 3(20+x)$ (1); $x = 20$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sum 3: 2 outcomes; sum 11: 2 outcomes (1); $P(3) = \frac{1}{18}$ (1); $P(11) = \frac{1}{18}$ (1); $P(\text{either}) = \frac{4}{36} = \frac{1}{9}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{6}{150} = \frac{1}{25}$ (1); Expected defective $= 3000 \times \frac{1}{25}$ (1); $= 120$ (1); Units stated (2).	1+1+1+2
20	Sum 7: 6 outcomes; sum 12: 1 outcomes (1); $P(7) = \frac{1}{6}$ (1); $P(12) = \frac{1}{36}$ (1); $P(\text{either}) = \frac{7}{36} = \frac{7}{36}$ (2).	1+1+1+2
	OR: $P(\text{defective}) = \frac{9}{300} = \frac{3}{100}$ (1); Expected defective $= 10000 \times \frac{3}{100}$ (1); $= 300$ (1); Units stated (2).	1+1+1+2

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- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
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